Ondansetron Kabi 2 mg/ml Solution for Injection

COMPOSITION

1 ml solution for injection contains:
Ondansetron hydrochloride dihydrate equivalent to 2 mg ondansetron.
Each ampoule with 2 ml contains 4 mg ondansetron.
Each ampoule with 4 ml contains 8 mg ondansetron.

1 ml solution for injection contains 3.34 mg of sodium as sodium citrate dihydrate and sodium chloride.

Full list of excipients:
Sodium chloride
Sodium citrate dihydrate
Citric acid monohydrate
Water for injections

PRODUCT DESCRIPTION

Solution for injection
Clear and colourless solution

PHARMACODYNAMIC PROPERTIES

Pharmacotherapeutic group: Antiemetics and antinauseants, Serotonin ($5HT_3$) antagonists
ATC Code: A04AA01

Ondansetron is a potent, highly selective $5HT_3$ receptor antagonist.
Its precise mode of action in the control of nausea and vomiting is not known. Chemotherapeutic agents and radiotherapy may cause release of 5HT in the small intestine initiating a vomiting reflex by activating vagal afferents via $5HT_3$ receptors. Ondansetron blocks the initiation of this reflex. Activation of vagal afferents may also cause a release of 5HT in the area postrema, located on the floor of the fourth ventricle, and this may also promote emesis through a central mechanism. Thus, the effect of ondansetron in the management of the nausea and vomiting induced by cytotoxic chemotherapy and radiotherapy is probably due to antagonism of $5HT_3$ receptors on neurons located both in the peripheral and central nervous system.

The mechanisms of action in post-operative nausea and vomiting are not known but there may be common pathways with cytotoxic induced nausea and vomiting.

Ondansetron does not alter plasma prolactin concentrations.

The role of ondansetron in opiate-induced emesis is not yet established.

PHARMACOKINETIC PROPERTIES

The pharmacokinetic properties of ondansetron are unchanged on repeat dosing.
A direct correlation of plasma concentration and anti-emetic effect has not been established.

Absorption
A 4 mg intravenous infusion of ondansetron given over 5 minutes results in peak plasma concentrations of about 65 ng/ml.

Distribution
The disposition of ondansetron following oral, intramuscular (IM) and intravenous (IV) dosing is similar with a steady state volume of distribution of about 140 L. Equivalent systemic exposure is achieved after IM and IV administration of ondansetron.
Ondansetron is not highly protein bound (70-76%).

Biotransformation
Ondansetron is cleared from the systemic circulation predominantly by hepatic metabolism through multiple enzymatic pathways. The absence of the enzyme CYP2D6 (the debrisoquine polymorphism) has no effect on ondansetron's pharmacokinetics.

Elimination
Less than 5% of the absorbed dose is excreted unchanged in the urine. Terminal half-life is about 3 hours.

Special Patient Populations

Gender differences

Gender differences were shown in the disposition of ondansetron, with females having a greater rate and extent of absorption following an oral dose and reduced systemic clearance and volume of distribution (adjusted for weight).

Children and Adolescents (aged 1 month to 17 years)

The half-life in the patient population aged 1 to 4 months was reported to average 6.7 hours compared to 2.9 hours for patients in the 5 to 24 months and 3 to 12-year age range. The differences in pharmacokinetic parameters in the 1 to 4 months patient population can be explained in part by the higher percentage of total body water in neonates and infants and a higher volume of distribution for water soluble drugs like ondansetron.

In paediatric patients aged 3 to 12 years undergoing elective surgery with general anaesthesia, the absolute values for both the clearance and volume of distribution of ondansetron were reduced in comparison to values with adult patients. Both parameters increased in a linear fashion with weight and by 12 years of age, the values were approaching those of young adults. When clearance and volume of distribution values were normalised by body weight, the values for these parameters were similar between the different age group populations. Use of weight-based dosing compensates for age-related changes and is effective in normalising systemic exposure in paediatric patients.

Elderly

Early Phase I studies in healthy elderly volunteers showed a slight age-related decrease in clearance, and an increase in half-life of ondansetron. However, wide inter-subject variability resulted in considerable overlap in pharmacokinetic parameters between young (< 65 years of age) and elderly subjects (≥ 65 years of age) and there were no overall differences in safety or efficacy observed between young and elderly cancer patients enrolled in CINV clinical trials to support a different dosing recommendation for the elderly. Based on more recent ondansetron plasma concentrations and exposure-response modelling, a greater effect on QTcF is predicted in patients ≥ 75 years of age compared to young adults. Specific dosing information is provided for patients over 65 years of age and over 75 years of age for IV dosing (see section *Posology and Method of Administration*).

Renal impairment

In patients with renal impairment (creatinine clearance 15-60 ml/min), both systemic clearance and volume of distribution are reduced following IV administration of ondansetron, resulting in a slight, but clinically insignificant, increase in elimination half-life (5.4 h). A study in patients with severe renal impairment who required regular haemodialysis (studied between dialyses) showed ondansetron's pharmacokinetics to be essentially unchanged following IV administration.

Hepatic impairment

In patients with severe hepatic impairment, ondansetron's systemic clearance is markedly reduced with prolonged elimination half-lives (15-32 h).

THERAPEUTIC INDICATIONS

Adults:

Ondansetron Kabi is indicated for management of nausea and vomiting induced by cytotoxic chemotherapy and radiotherapy

Ondansetron Kabi is also indicated for the prevention and treatment of post-operative nausea and vomiting (PONV).

Paediatric Population:

Ondansetron Kabi is indicated for the management of nausea and vomiting induced by cytotoxic chemotherapy.

POSOLOGY AND METHOD OF ADMINISTRATION

For intravenous injection or for intravenous infusion after dilution.
For instructions on dilution of the product before administration, see section *Special precautions for storage.*

<u>Chemotherapy and radiotherapy induced nausea and vomiting (CINV and RINV)</u>

Adults:

The emetogenic potential of cancer treatment varies according to the doses and combinations of chemotherapy and radiotherapy regimens used. The route of administration and dose of Ondansetron Kabi should be flexible in the range of 8-32 mg a day and selected as shown below.

Emetogenic chemotherapy and radiotherapy

For patients receiving emetogenic chemotherapy or radiotherapy ondansetron can be given either by intravenous or oral administration.
The recommended intravenous dose of ondansetron is 8 mg administered as a slow intravenous injection in not less than 30 seconds immediately before treatment.
Oral or rectal treatment is recommended to protect against delayed or prolonged emesis after the first 24 hours.

For oral or rectal administration refer to the SmPC of ondansetron tablets and suppositories, respectively.

Highly emetogenic chemotherapy e.g. high-dose cisplatin
Ondansetron Kabi may be administered as a single 8 mg intravenous dose immediately before chemotherapy. IV doses greater than 8 mg and up to a maximum of 16 mg of ondansetron may only be given by intravenous infusion diluted in 50 mL to 100 mL of 0.9% Sodium Chloride Injection or 5% Dextrose Injection before administration and infused over not less than 15 minutes. A single dose greater than 16 mg must not be given due to dose dependent increase of QT-prolongation risk (see sections *Special warnings and precautions for use, Undesirable effects* and *Pharmacodynamic properties).*

For management of highly emetogenic chemotherapy, a dose of 8 mg may be administered by slow intravenous injection in not less than 30 seconds, followed by two further intravenous doses of 8 mg four hours apart, or by a constant infusion of 1 mg/hour for up to 24 hours.

The efficacy of Ondansetron Kabi in highly emetogenic chemotherapy may be enhanced by the addition of a single intravenous dose of dexamethasone sodium phosphate, 20 mg administered prior to chemotherapy.

Oral or rectal treatment is recommended to protect against delayed or prolonged emesis after the first 24 hours.
For oral or rectal administration refer to the SmPC of ondansetron tablets and suppositories, respectively.

Paediatric Population

CINV in children aged ≥6 months and adolescents:
The dose for CINV can be calculated based on body surface area (BSA) or weight – see below. In paediatric clinical studies, ondansetron was given by intravenous infusion diluted in 25 to 50 ml of saline or other compatible infusion fluid (see section 6.6) and infused over not less than 15 minutes.

Weight-based dosing results in higher total daily doses compared to BSA-based dosing (see sections 4.4)

Ondansetron Kabi should be diluted in 5% dextrose or 0.9% sodium chloride or other compatible infusion fluid (see section 6.6.) and infused intravenously over not less than 15 minutes.

Dosing by BSA:
Ondansetron Kabi should be administered immediately before chemotherapy as a single intravenous dose of 5 mg/m^2. The intravenous dose must not exceed 8 mg.
Oral dosing can commence 12 hours later and may be continued for up to 5 days. See Table 1 below.
The total dose over 24 hours (given as divided doses) must not exceed adult dose of 32 mg.

Table 1: BSA-based dosing for Chemotherapy induced nausea and vomiting - Children aged ≥6 months and adolescents[a]

BSA	Day 1[b,c]	Days 2-6[c]
< 0.6 m^2	5 mg/m^2 i.v. plus 2 mg syrup after 12 hours	2 mg syrup every 12 hours
≥ 0.6 m^2 to ≤ 1.2 m^2	5 mg/m^2 i.v. plus 4 mg syrup or tablet after 12 hours	4 mg syrup or tablet every 12 hours
>1.2 m^2	5 mg/m^2 or 8 mg i.v. plus 8 mg syrup or tablet after 12 hours	8 mg syrup or tablet every 12 hours

[a] Not all pharmaceutical forms may be available.
[b] The intravenous dose must not exceed 8 mg.
[c] The total dose over 24 hours (given as divided doses) must not exceed adult dose of 32 mg.

Dosing by bodyweight:
Weight-based dosing results in higher total daily doses compared to BSA-based dosing (see sections 4.4)
Ondansetron Kabi should be administered immediately before chemotherapy as a single intravenous dose of 0.15 mg/kg. The single intravenous dose must not exceed 8 mg.
Two further intravenous doses may be given in 4-hourly intervals. The total dose over 24 hours (given as divided doses) must not exceed adult dose of 32 mg.
Oral dosing can commence twelve hours later and may be continued for up to 5 days. See Table 2 below.

Table 2: Weight-based dosing for Chemotherapy induced nausea and vomiting - Children aged ≥6 months and adolescents[a]

Weigth	Day 1[b,c]	Days 2-6[c]
≤ 10 kg	Up to 3 doses of 0.15 mg/kg i.v. every 4 hours	2 mg syrup every 12 hours
> 10 kg	Up to 3 doses of 0.15 mg/kg i.v. every 4 hours	4 mg syrup or tablet every 12 hours

[a] Not all pharmaceutical forms may be available.
[b] The intravenous dose must not exceed 8 mg.
[c] The total dose over 24 hours (given as divided doses) must not exceed adult dose of 32 mg.

Elderly:

Ondansetron is well tolerated by patients over 65 years of age.

In patients 65 to 74 years of age, the dose schedule for adults can be followed. All intravenous doses should be diluted in 50-100 ml of saline or other compatible infusion fluid (see *Pharmaceutical Precautions*) and infused over 15 minutes.

In patients 75 years of age or older, the initial intravenous dose of Ondansetron Kabi should not exceed 8 mg. All intravenous doses should be diluted in 50-100 ml of saline or other compatible infusion fluid (see *Pharmaceutical Precautions*) and infused over 15 minutes.

The initial dose of 8 mg may be followed by 2 further intravenous doses of 8 mg, infused over 15 minutes and given no less than 4 hours apart (see *Pharmacokinetic Properties).*

Patients with renal impairment
No alteration of daily dosage or frequency of dosing, or route of administration are required.

Patients with hepatic impairment
Clearance of Ondansetron Kabi is significantly reduced and serum half-life significantly prolonged in subjects with moderate or severe impairment of hepatic function. In such patients, a total daily dose of 8 mg should not be exceeded.

Patients with poor sparteine/debrisoquine metabolism
The elimination half-life of ondansetron is not altered in subjects classified as poor metabolisers of sparteine and debrisoquine. Consequently, in such patients repeat dosing will give drug exposure levels no different from those of the general population. No alteration of daily dosage or frequency of dosing are required.

Post-operative nausea and vomiting (PONV)

Adults:
Prevention of PONV
For prevention of post-operative nausea and vomiting, the recommended dose of Ondansetron Kabi injection is a single dose of 4 mg by slow intravenous injection in not less than 30 seconds administered at the induction of anaesthesia.

Treatment of established PONV
For treatment of established PONV a single dose of 4 mg given by slow intravenous injection in not less than 30 seconds is recommended.

Paediatric population:
Post-operative nausea and vomiting in children aged ≥1 month and adolescents

For prevention of PONV in paediatric patients having surgery performed under general anaesthesia, a single dose of Ondansetron Kabi may be administered by slow intravenous injection (not less than 30 seconds) at a dose 0.1 mg/kg up to a maximum of 4 mg either prior to, at or after induction of anaesthesia.

For the treatment of PONV after surgery in paediatric patients having surgery performed under general anaesthesia, a single dose of Ondansetron Kabi may be administered by slow intravenous injection (not less than 30 seconds) at a dose of 0.1 mg/kg up to a maximum of 4 mg.
There are no data on the use of Ondansetron Kabi for treatment of postoperative vomiting in children below 2 years of age.

Elderly:
There is limited experience in the use of Ondansetron Kabi in the prevention and treatment of post-operative nausea and vomiting (PONV) in older people, however ondansetron is well tolerated in patients over 65 years receiving chemotherapy.

Special Populations

Patients with renal impairment
No alteration of daily dosage or frequency of dosing, or route of administration is required.

Patients with hepatic impairment
Clearance of Ondansetron Kabi is significantly reduced and serum half-life significantly prolonged in subjects with moderate or severe impairment of hepatic function. In such patients a total daily dose of 8 mg (orally or parenterally) should not be exceeded.

Patients with poor sparteine/debrisoquine metabolism
The elimination half-life of ondansetron is not altered in subjects classified as poor metabolizers of sparteine and debrisoquine. Consequently, in such patients repeat dosing will give drug exposure levels no different from those of the general population. No alteration of daily dosage or frequency of dosing is required.

CONTRAINDICATIONS

Concomitant use with apomorphine (see *Interactions with other medicinal products*).

Hypersensitivity to any component of the preparation.

SPECIAL WARNINGS AND PRECAUTIONS FOR USE

Hypersensitivity reactions have been reported in patients who have exhibited hypersensitivity to other selective 5HT3 receptor antagonists.

Respiratory events should be treated symptomatically, and clinicians should pay particular attention to them as precursors of hypersensitivity reactions.

Ondansetron prolongs the QT interval in a dose-dependent manner (see Pharmacodynamic properties). In addition, post-marketing cases of Torsade de Pointes have been reported in patients using ondansetron. Avoid ondansetron in patients with congenital long QT syndrome. Ondansetron should be administered with caution to patients who have or may develop prolongation of QTc, including patients with electrolyte abnormalities, congestive heart failure, bradyarrhythmias or patients taking other medicinal products that lead to QT prolongation or electrolyte abnormalities.

Myocardial ischemia have been reported in patients treated with ondansetron. In some cases, predominantly during intravenous administration, the symptoms appeared immediately after administration but recovered with prompt treatment. Therefore, caution should be exercised during and after administration of ondansetron.

Hypokalemia and hypomagnesemia should be corrected prior to ondansetron administration.

There have been post-marketing reports describing patients with serotonin syndrome (including altered mental status, autonomic instability and neuromuscular abnormalities) following the concomitant use of ondansetron and other serotonergic drugs (including selective serotonin reuptake inhibitors (SSRI) and serotonin noradrenaline reuptake inhibitors (SNRIs)). If concomitant treatment with ondansetron and other serotonergic drugs is clinically warranted, appropriate observation of the patient is advised.
As ondansetron is known to increase large bowel transit time, patients with signs of sub-acute intestinal obstruction should be monitored following administration.

In patients with adenotonsillar surgery prevention of nausea and vomiting with ondansetron may mask occult bleeding. Therefore, such patients should be followed carefully after ondansetron.
This medicinal product contains 2.3 mmol (or 53.5 mg) sodium per dose. To be taken into consideration by patients on a controlled sodium diet.

Paediatric Population:

Paediatric patients receiving ondansetron with hepatotoxic chemotherapeutic agents should be monitored closely for impaired hepatic function.

CINV
When calculating the dose on an mg/kg basis and administering three doses at 4-hour intervals, the total daily dose will be higher than if one single dose of 5 mg/m^2 followed by an oral dose is given. The comparative efficacy of these two different dosing regimes has not been investigated in clinical trials. Cross trial comparing indicate similar efficacy for both regimes (see section *Pharmacodynamic properties*)

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

In psychomotor testing ondansetron does not impair performance nor cause sedation. No detrimental effects on such activities are predicted from the pharmacology of ondansetron.

INTERACTION WITH OTHER MEDICINAL PRODUCTS AND OTHER FORMS OF INTERACTION

There is no evidence that ondansetron either induces or inhibits the metabolism of other drugs commonly coadministered with it. Specific studies have shown that there are no interactions when ondansetron is administered with alcohol, temazepam, furosemide, alfentanil, tramadol, morphine, lidocaine, thiopental or propofol.

Ondansetron is metabolised by multiple hepatic cytochrome P-450 enzymes: CYP3A4, CYP2D6 and CYP1A2. Due to the multiplicity of metabolic enzymes capable of metabolising ondansetron, enzyme inhibition or reduced activity of one enzyme (e.g.

CYP2D6 genetic deficiency) is normally compensated by other enzymes and should result in little or no significant change in overall ondansetron clearance or dose requirement.

Caution should be exercised when ondansetron is coadministered with drugs that prolong the QT interval and/or cause electrolyte abnormalities. (See *Special warnings and precautions for use*).

Use of Ondansetron with QT prolonging drugs may result in additional QT prolongation. Concomitant use of ondansetron with cardiotoxic drugs (e.g. anthracyclines (such as doxorubicin, daunorubicin) or trastuzimab), antibiotics (such as erythromycin), antifungals (such as ketoconazole), antiarrhythmics (such as amiodarone) and beta blockers (such as atenolol or timolol) may increase the risk of arrhythmias. (See *Special warnings and precautions for use*).

There have been post-marketing reports describing patients with serotonin syndrome (including altered mental status, autonomic instability and neuromuscular abnormalities) following the concomitant use of ondansetron and other serotonergic drugs (including SSRIs and SNRIs). (See *Special warnings and precautions for use*)

Apomorphine
Based on reports of profound hypotension and loss of consciousness when ondansetron was administered with apomorphine hydrochloride, concomitant use with apomorphine is contraindicated.

Phenytoin, Carbamazepine and Rifampicin
In patients treated with potent inducers of CYP3A4 (i.e. phenytoin, carbamazepine, and rifampicin), the oral clearance of ondansetron was increased and ondansetron blood concentrations were decreased.

Tramadol
Data from small studies indicate that ondansetron may reduce the analgesic effect of tramadol.

FERTILITY, PREGNANCY AND LACTATION

<u>Pregnancy and breast-feeding</u>
The use of Ondansetron in pregnancy is not recommended.

In human epidemiological studies, an increase in orofacial clefts was observed in infants of women administered Ondansetron during the first trimester of pregnancy. Regarding cardia malformations, the epidemiological studies showed confliction results.

Three epidemiological studies in the US assessed the risk of specific congenital anomalies, including orofacial clefts and cardia malformations in offspring born to mothers exposed to Ondansetron during the first trimester of pregnancy.

- One cohort study with 88,467 pregnancies exposed to Ondansetron showed and increased risk of oral clefts (3 additional cases per 10,000 women treated, adjusted relative risk (RR), 1.24% (95% CI 1.03-1.48)) without an apparent increase in risk of cardiac malformations. A separately published subgroup analysis of 23,877 pregnancies exposed to intravenous Ondansetron did not find and increased risk of either oral clefts or cardiac malformations.

- One case-control study using population-based birth defect registries with 23,200 cases across two datasets showed and increased risk of cleft palate in one dataset and no increased risk in the other dataset.
- The second cohort study with 3,733 pregnancies exposed to Ondansetron found an increased risk of ventricular septal defect, adjusted RR 1.7 (95%CI 1.0-2.9), but no statistically significant increase in risk of cardiac malformations.

Reproductive studies in rats and rabbits did not show evidence of harm to the fetus.

Pregnancy status should be verified for females of reproductive potential prior to starting the treatment with Ondansetron Kabi 2 mg/ml Solution for Injection.

Females of reproductive potential should be advised that it is possible that Ondansetron Kabi 2 mg/ml Solution for Injection can cause harm to the developing fetus. Sexually active females of reproductive potential are recommended to use effective contraception (methods that result in less than 1% pregnancy rates) when using Ondansetron Kabi 2 mg/ml Solution for Injection during the treatment and for two days after stopping treatment with Ondansetron Kabi 2 mg/ml Solution for Injection.

UNDESIRABLE EFFECTS

Adverse events are listed below by system organ class and frequency. Frequencies are defined as: very common, common, uncommon, rare, very rare and not known.

The following frequencies are estimated at the standard recommended doses of ondansetron according to indication and formulation. The adverse event profiles in children and adolescents were comparable to that seen in adults.

Very Common	**Common**	**Uncommon**	**Rare**	**Very rare**	**Not known**
Immune system disorders					
			Immediate hypersensitivity reactions sometimes severe, including anaphylaxis[1].		
Nervous system disorders					
Headache.		Seizures, movement disorders (including extrapyramidal reactions such as dystonic reactions,	Dizziness during rapid IV administration.		

Very Common	Common	Uncommon	Rare	Very rare	Not known
		oculogyric crisis and dyskinesia)[2]			
Eye disorders					
			Transient visual disturbances (e.g. blurred vision) predominantly during IV administration.	Transient blindness predominantly during intravenous administration[3].	
Cardiac disorders					
		Arrhythmias, chest pain with or without ST segment depression, bradycardia.	QTc prolongation (including Torsade de pointes)		Myocardial ischemia
Vascular disorders					
	Sensation of warmth or flushing.	Hypotension.			
Respiratory, thoracic and mediastinal disorders					
		Hiccups.			
Gastrointestinal disorders					
	Constipation.				
Hepatobiliary disorders					
		Asymptomatic increases in liver function tests[4]			
Skin and subcutaneous tissue disorders					
				Toxic skin eruption	

Very Common	Common	Uncommon	Rare	Very rare	Not known
				(including toxic epidermal necrolysis)	
General disorders and administration site conditions					
	Local IV injection site reactions – in particular by repeated administration				

[1.] Anaphylaxis can be life-threatening. Hypersensitivity reactions were also observed in patients who have shown these symptoms with other selective $5HT_3$ reseptor antagonist.

[2]. Observed without definitive evidence of persistent clinical sequelae.

[3.] The majority of the blindness cases reported resolved within 20 minutes. Most patients had received chemotherapeutic agents, which included cisplatin. Some cases of transient blindness were reported as cortical in origin.

[4]. These events were observed commonly in patients receiving chemotherapy with cisplatin.

OVERDOSE
Symptoms and Signs

There is limited experience of ondansetron overdose. In the majority of cases, symptoms were similar to those already reported in patients receiving recommended doses (see *Undesirable effects*). Manifestations that have been reported include visual disturbances, severe constipation, hypotension and a vasovagal episode with transient second-degree AV block.

Ondansetron prolongs QT interval in a dose-dependent manner. ECG monitoring is recommended in cases of overdose.

Paediatric population

Paediatric cases consistent with serotonin syndrome have been reported after inadvertent oral overdoses of ondansetron (exceeded estimated ingestion of 4 mg/kg) in infants and children aged 12 months to 2 years.

Treatment

There is no specific antidote for ondansetron, therefore in cases of suspected overdose, symptomatic and supportive therapy should be given as appropriate.

The use of ipecacuanha to treat overdose with ondansetron is not recommended, as patients are unlikely to respond due to the anti-emetic action of ondansetron itself.

INCOMPATIBILITIES

Ondansetron Kabi injection should not be administered in the same syringe or infusion as any other medication.

COMPATIBILITIES

Ondansetron Kabi injection should only be mixed with those infusion solutions that are recommended in section *Pharmaceutical precautions.*

PHARMACEUTICAL PRECAUTIONS

Any unused product or waste material should be disposed of in accordance with local requirements.

Ondansetron Kabi 2 mg/ml may be diluted with the following solutions for infusion:
Sodium chloride 9 mg/ml (0.9 % w/v) solution
Glucose 50 mg/ml (5 % w/v) solution
Mannitol 100 mg/ml (10 % w/v) solution
Ringer's lactate solution

Method of Dilution:

For intravenous infusion used for adults the product should be diluted in 50 mL to 100 mL and when used for paediatric population then it should be diluted in diluted in 25 to 50 ml.

The diluted solutions should be stored protected from light.

Note:
Ondansetron Kabi injection ampoules should not be autoclave

STORAGE

Store below 30°C.
Keep the ampoules in the outer carton in order to protect from light.
For storage conditions of the diluted medicinal product,
Infusion: Chemical and physical in-use stability has been demonstrated for 48 hours at 25°C with the solutions given in section " Special precautions for disposal and other handling of the product."
From a microbiological point of view, the product should be used immediately. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user and would normally not be longer than 24 hours at 2 to 8°C, unless dilution has taken place in controlled and validated aseptic conditions.
The diluted solutions should be stored protected from light.

SHELF-LIFE

a) Unopened: 4 years

b) Injection: After first opening the medicinal product should be used immediately.

c) Infusion: Chemical and physical in-use stability has been demonstrated for 48 hours at 25°C with the solutions given in section "Special precautions for disposal and other handling of the product."

From a microbiological point of view, the product should be used immediately. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user and would normally not be longer than 24 hours at 2 to 8°C, unless dilution has taken place in controlled and validated aseptic conditions.

PRESENTATION

Type I clear glass ampoules

2 ml:

Pack sizes: 1, 5 and 10 ampoules.

4 ml:

Pack sizes: 1, 5 and 10 ampoules.

Not all pack sizes may be marketed.

MANUFACTURED BY:

Labesfal Laboratorios Almiro S.A.
Fresenius Kabi Group
3465-157 Santiago de Besteiros
Portugal

PRODUCT REGISTRATION HOLDER/IMPORTER:

Fresenius Kabi Malaysia Sdn Bhd
3-1 & 3-2, Jalan 51A/225, 46100 Petaling Jaya,
Selangor, Malaysia

REVISION DATE

July 2022

FRESENIUS KABI